fabric
FEBRUARY

Masterclass in Event-driven Architecture with Microsoft Fabric

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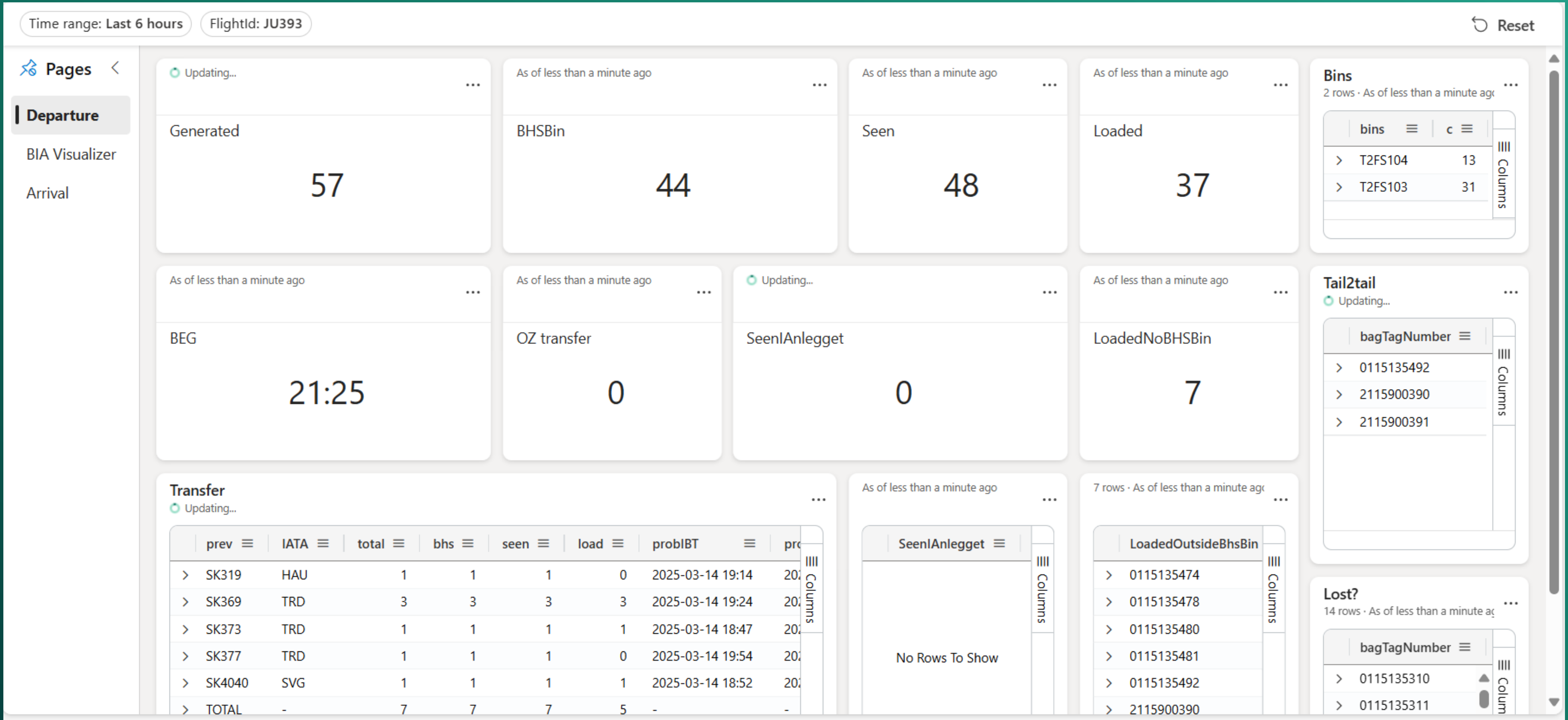
6 February 2026

Real-Time Baggage Handling



What can you expect today?

- Dive into event-driven architectures and product capabilities
- Use cases and examples
- Hands-on lab

What we expect from you?

- Be curious, ask questions (raise your hand)
- Follow the instructions for hands-on lab
- Be kind to your neighbours, help each other to learn

Today’s agenda

08:30 – 10:00	Welcome and introduction Event-driven Architectures – Why & What? Setting up the lab environment
10:00 – 10:20	Morning Break
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Event-driven Architectures and Patterns

Happening across a variety of industries

- Airlines
- Manufacturing
- Energy
- Retail
- Health Care
- Logistics
- Airports
- Banking
- Hospitality
- Telecoms
- Sports
- Automotive

Common business and operational needs today

- I want to know when something changes in my operational systems (database)
- I want continuous, granular insights from my assets
- I want near real-time reporting, ML scoring and GenAI apps
- I want to process/act/react on events occurring in my physical or digital asset
- I want real-time view into customer interactions and transactions

Pattern practiced and perfected over time

Ingest & process

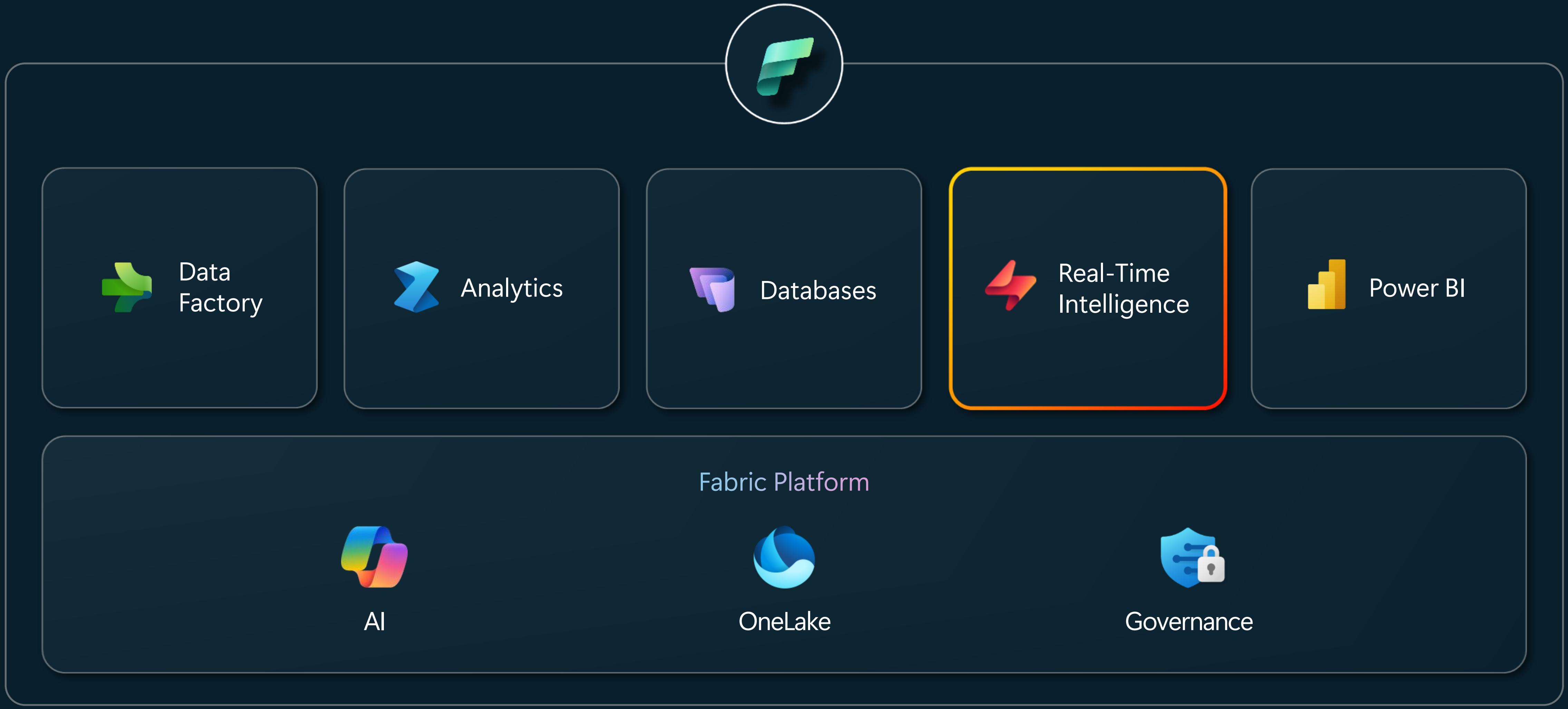
Analyze & transform

Visualize

Batch Sources

Microsoft Fabric

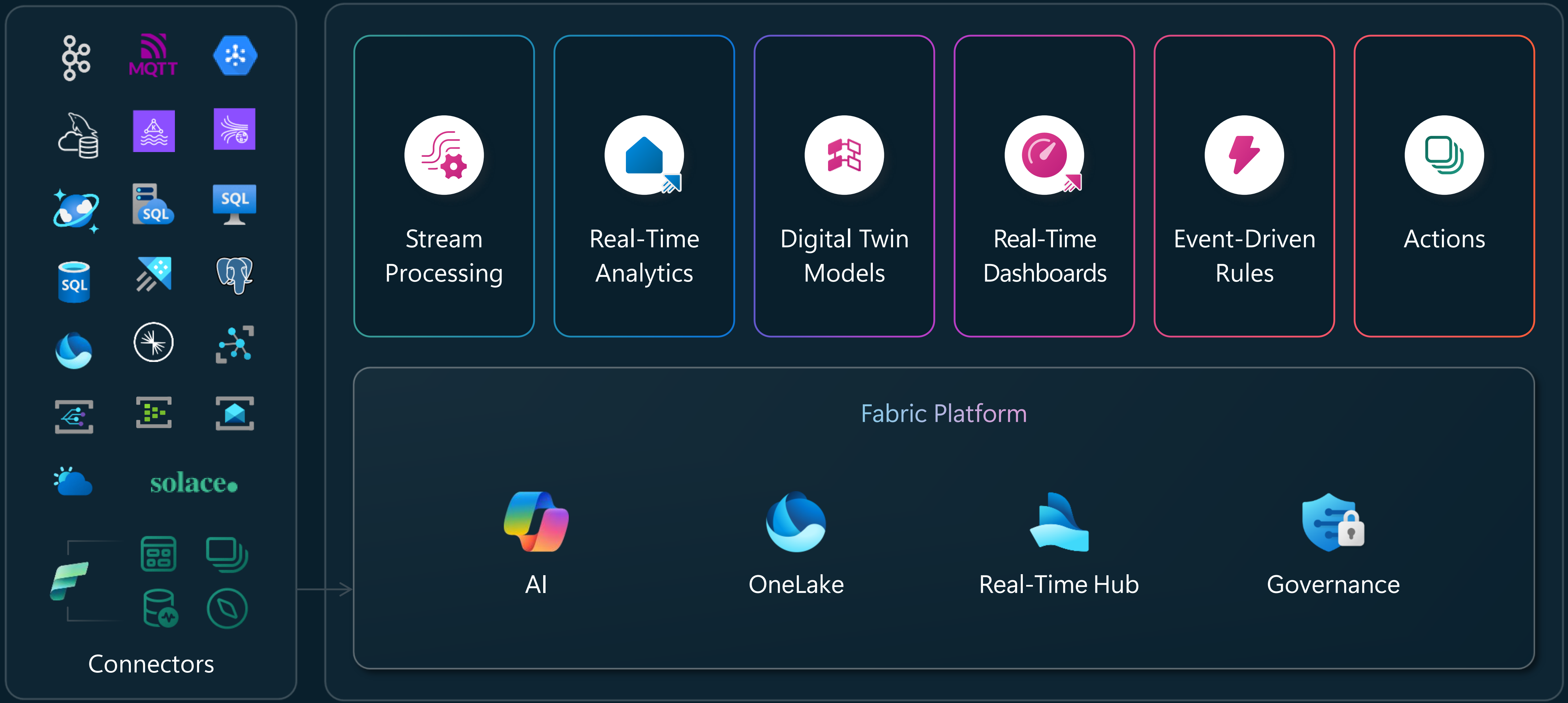
The unified data platform for AI transformation



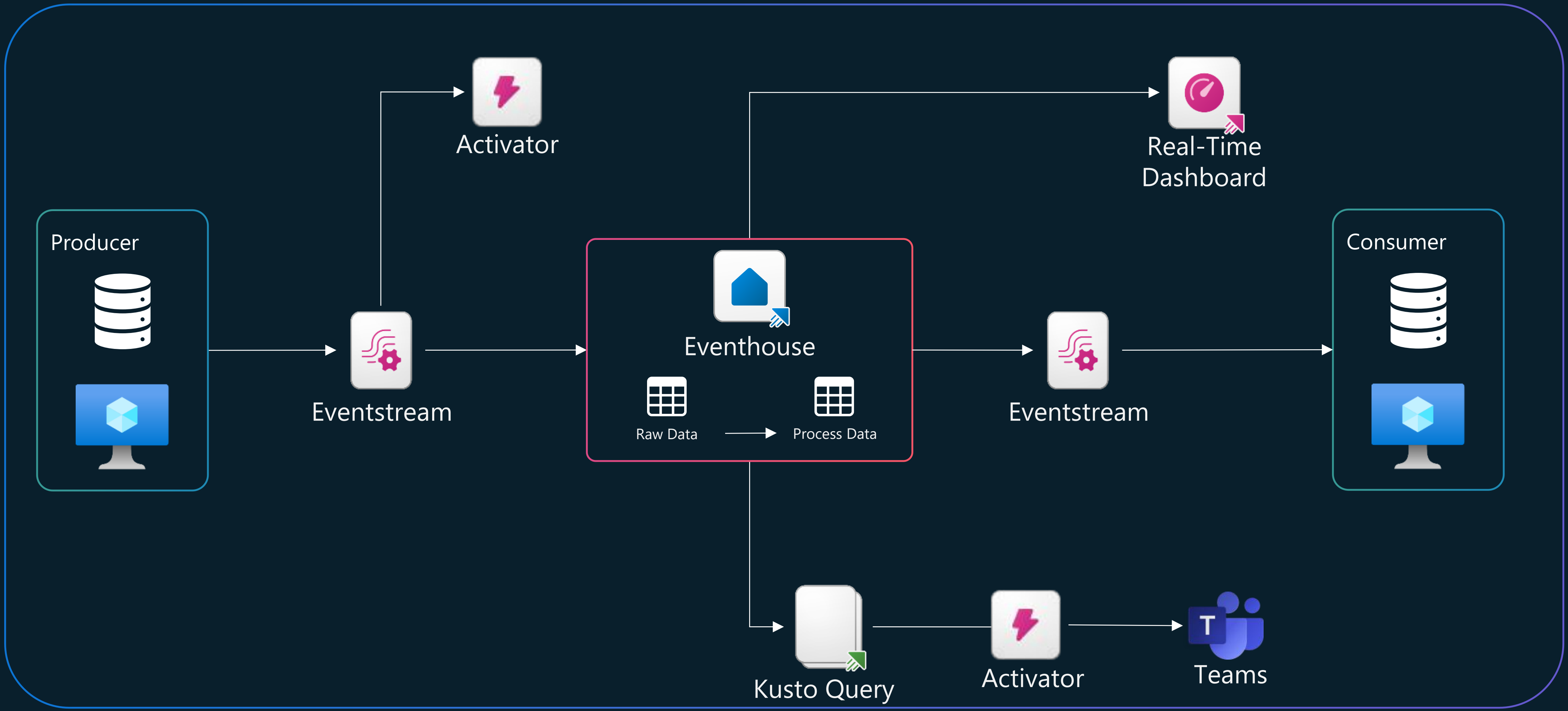
Real-Time Intelligence in Microsoft Fabric

Enterprise real-time data platforms	+	Self-serve experiences for business users	+	Intelligent insights and capabilities	=	Real-Time Intelligence in Microsoft Fabric
Azure Event Hubs		Power BI		AI skills		Fully integrated
Azure Event Grid		Activator		Anomaly detection		SaaS-native experiences
Azure Stream Analytics		Real-Time Hub		Agents		Single data estate
Azure Data Explorer		OneLake				Unified data platform

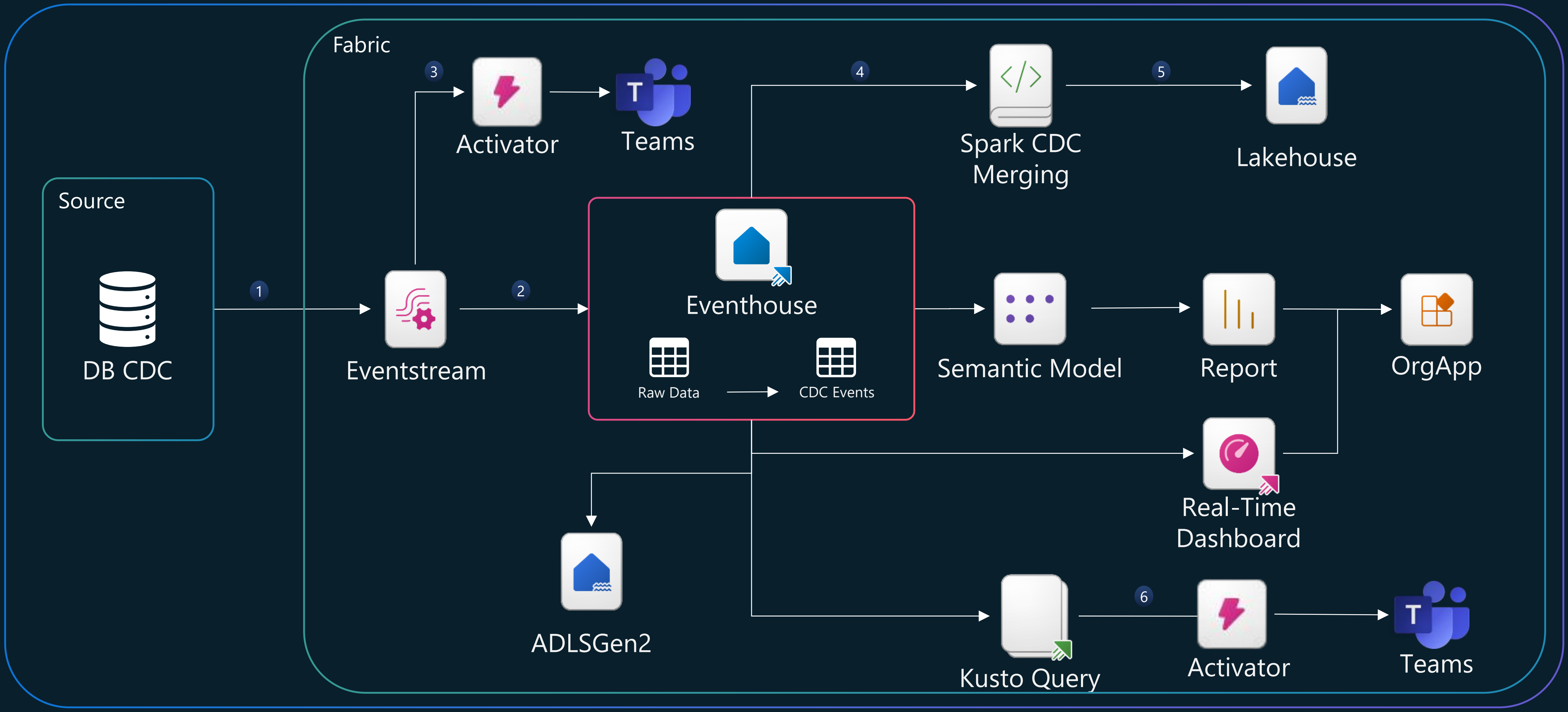
Real-Time Intelligence in Microsoft Fabric

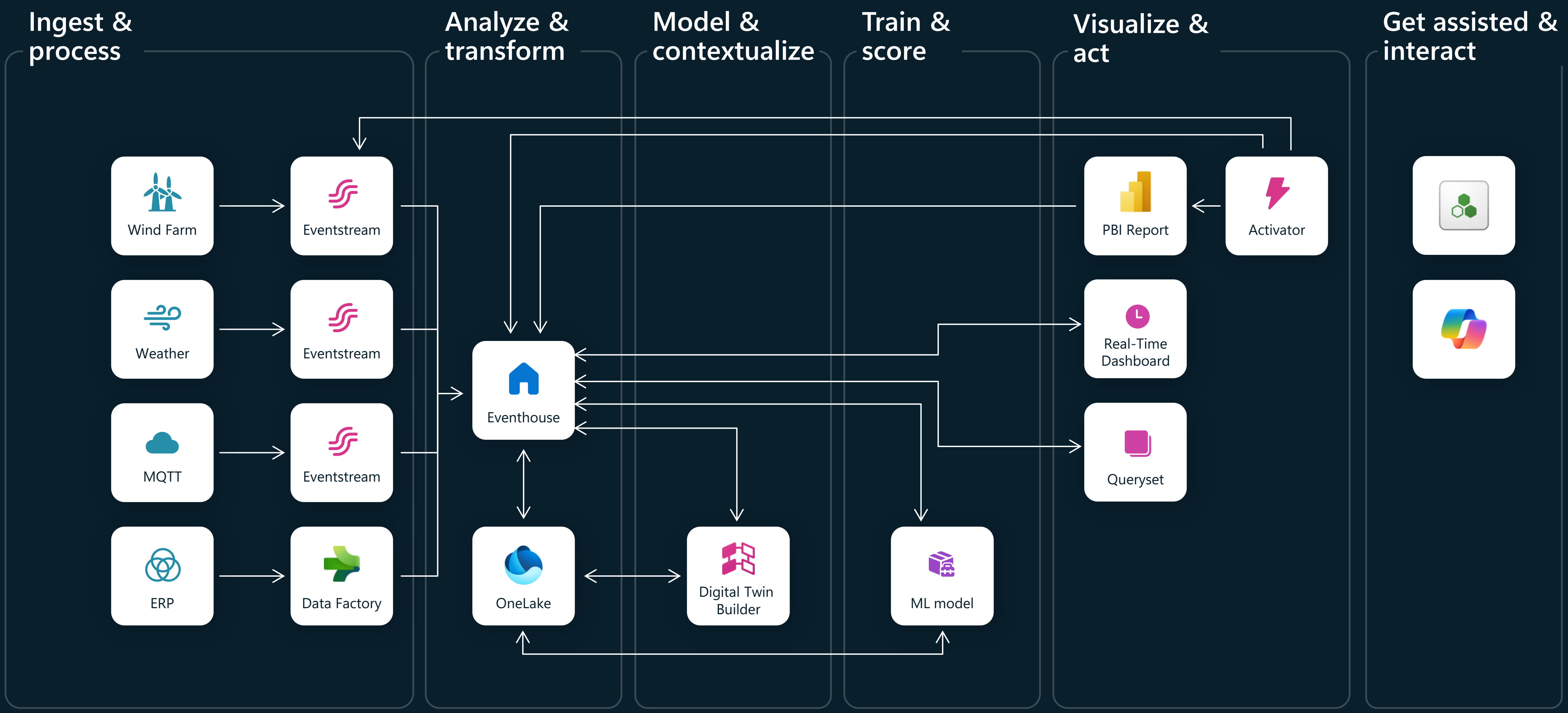


Pub-Sub Stream Processor



Database Change Data Capture Processing



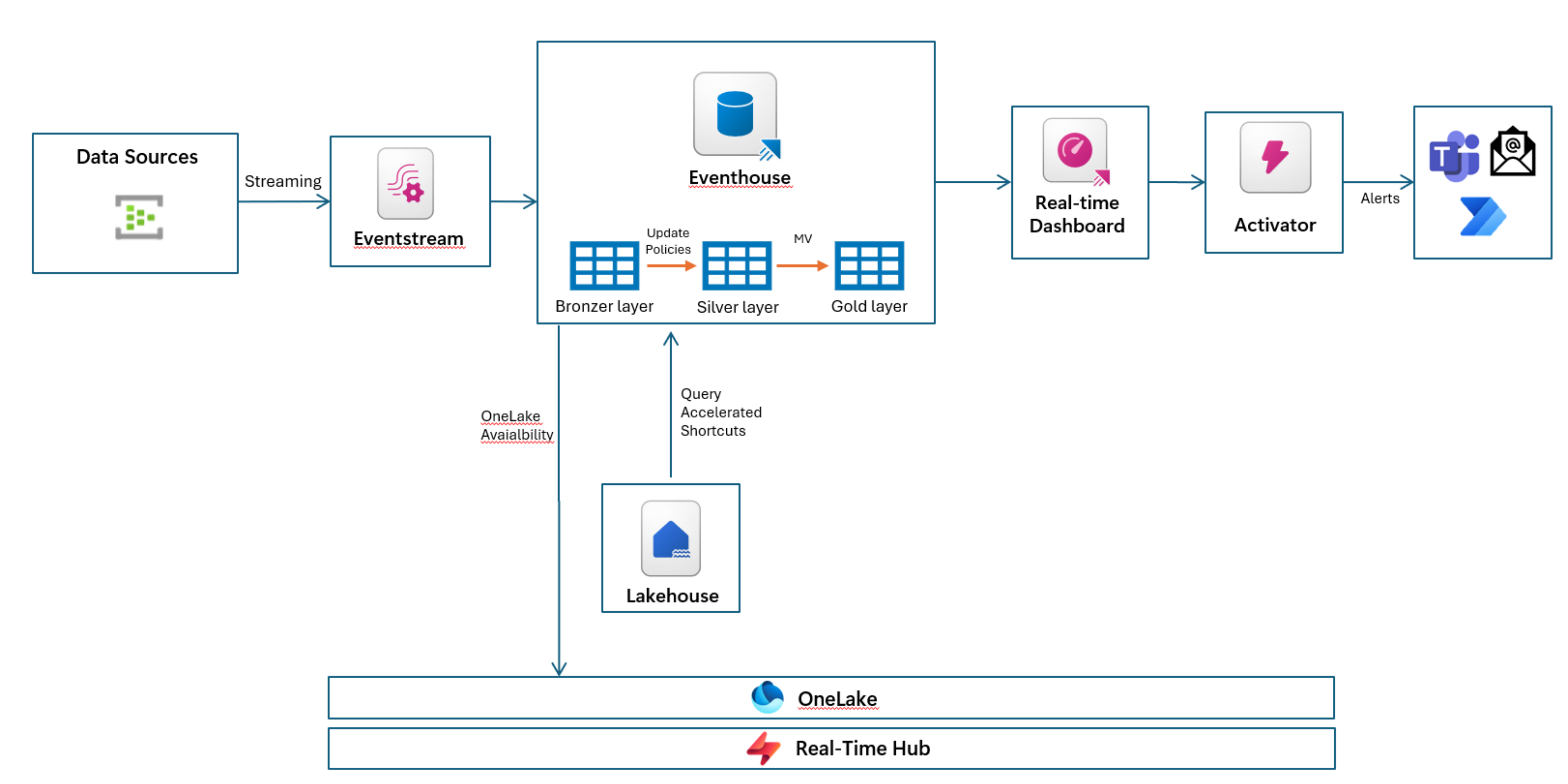

Ingest & process
Analyze & transform
Model & contextualize
Train & score
Visualize & act
Get assisted & interact
Wind Farm
Eventstream
Weather
Eventstream
MQTT
Eventstream
ERP
Data Factory
Eventhouse
OneLake
Digital Twin Builder
ML model
PBI Report
Activator
Real-Time Dashboard
Queryset

Setting up the lab

The full lab instructions

https://aka.bi/rtilab

Workshop Architecture



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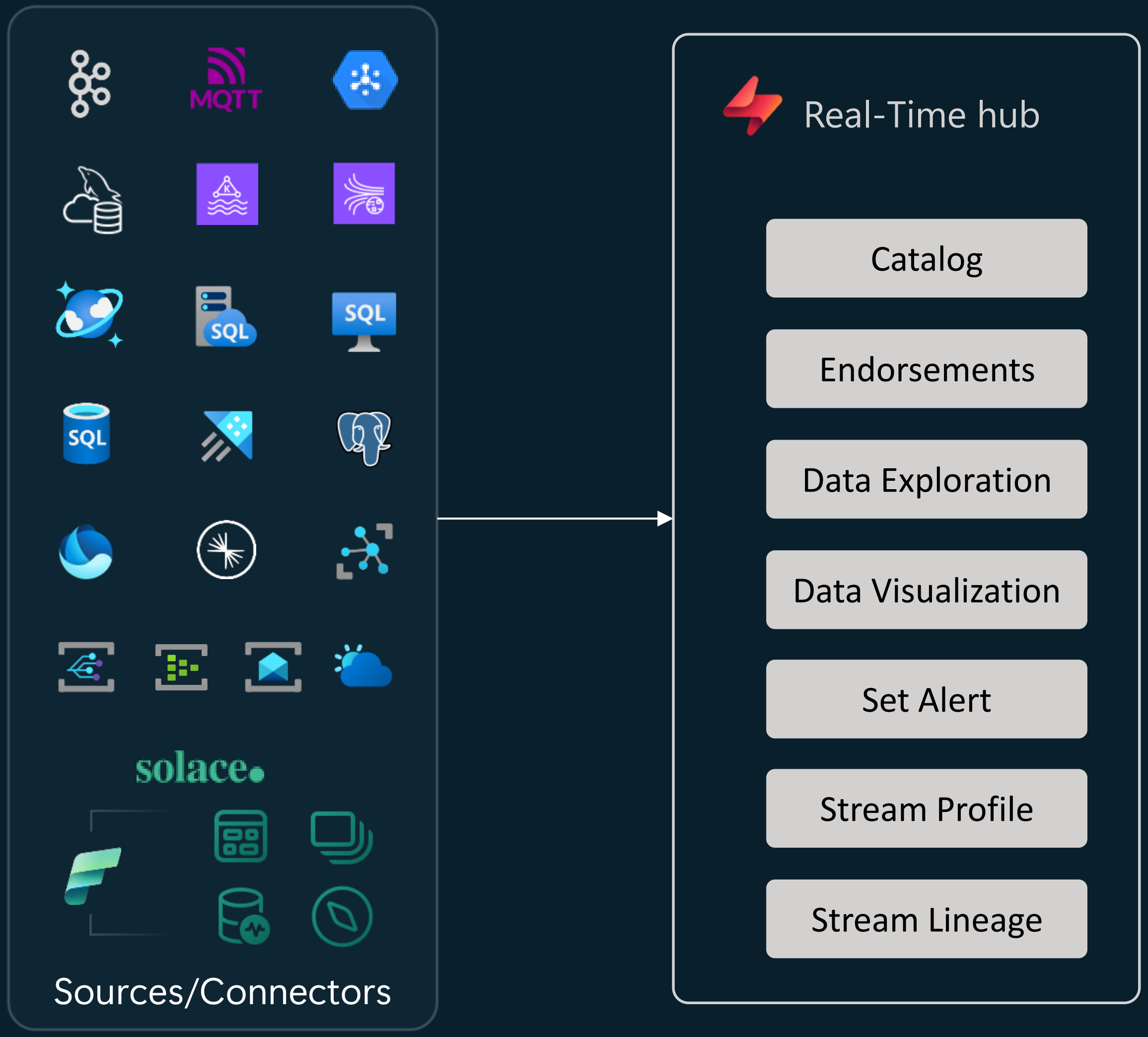
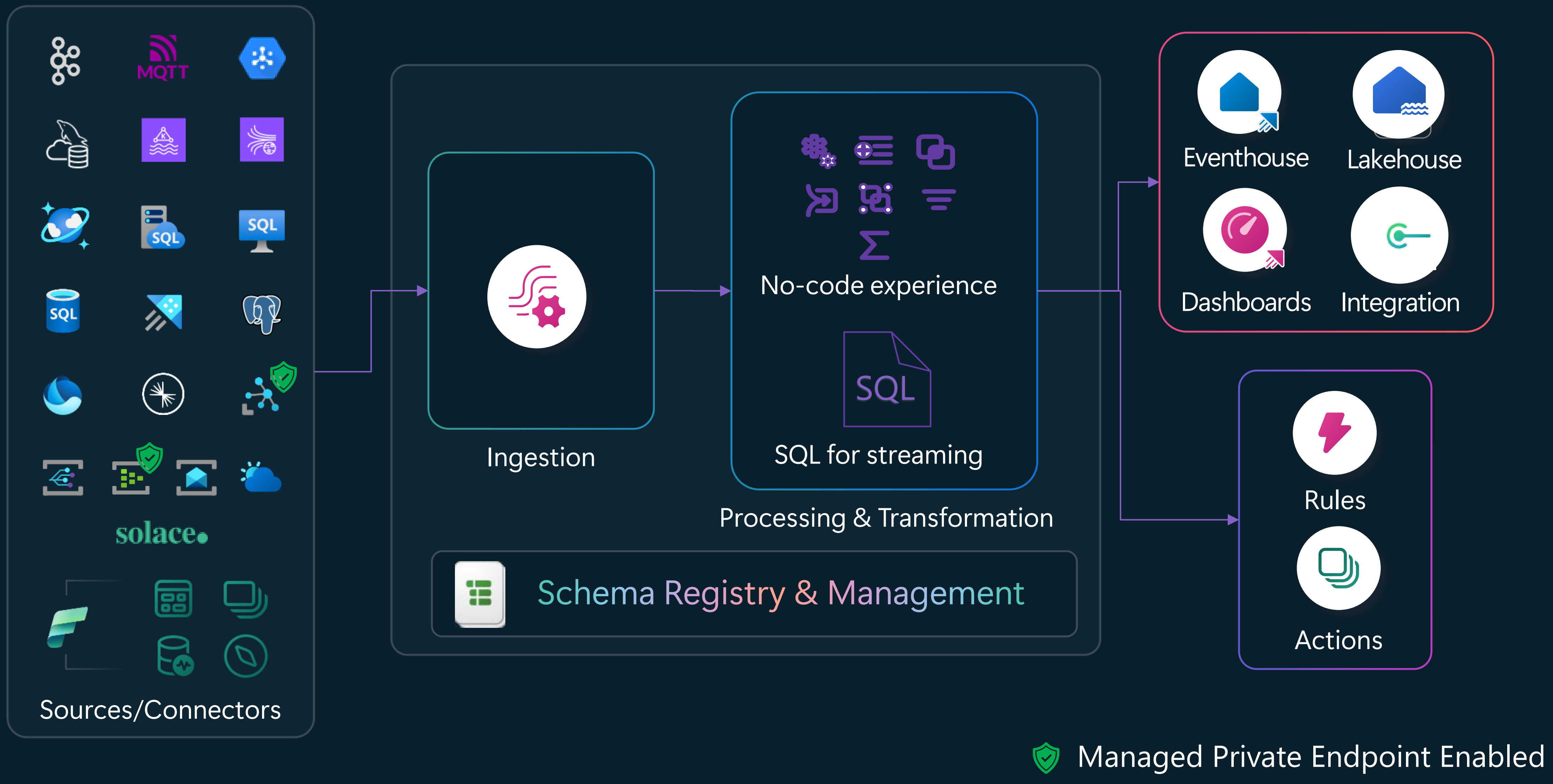
Real-Time hub



Ingest, process, and route events using Eventstream



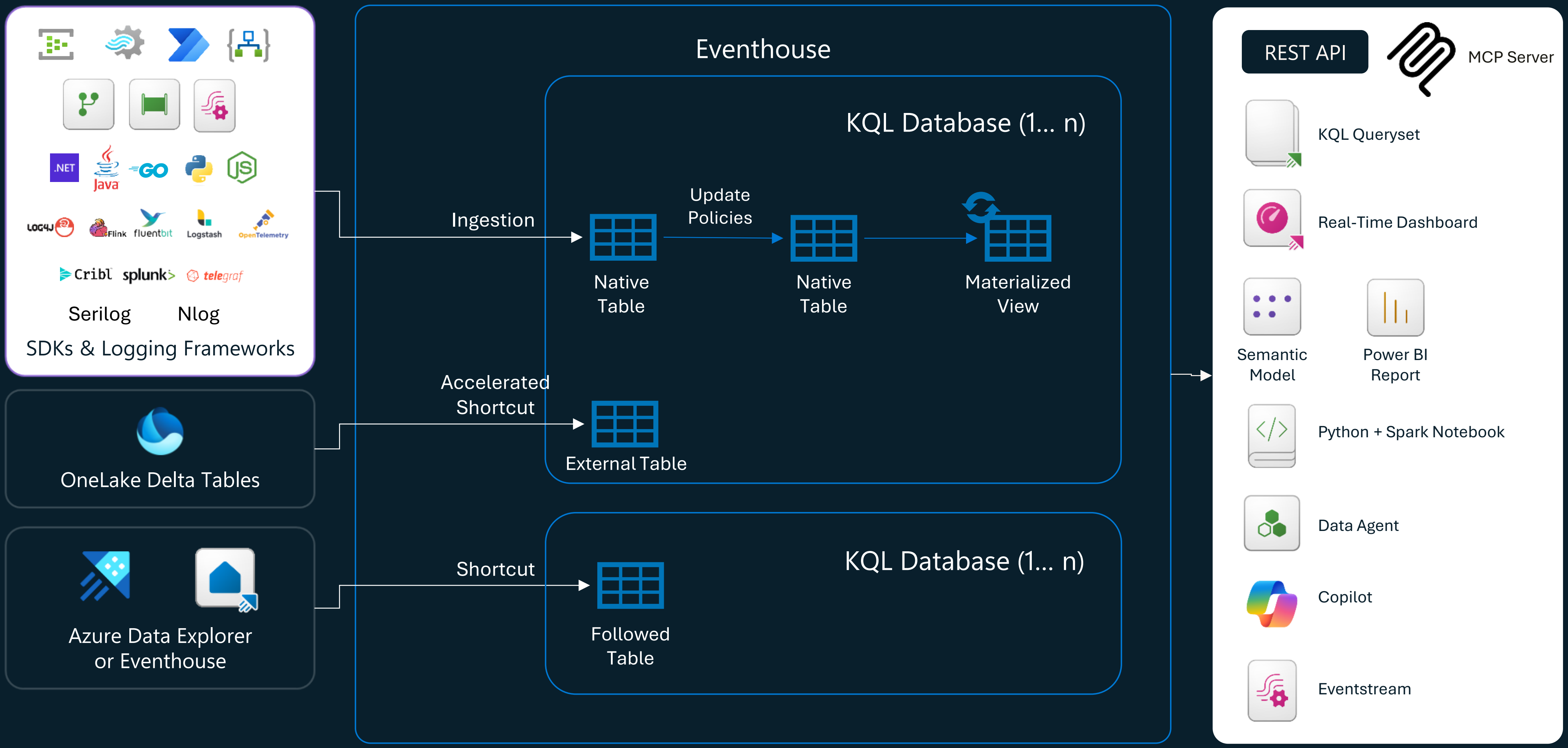
Connect events from many types of sources

Examples of connectors and sources

Change Data Capture (CDC)	CDC for Azure SQL DB	CDC for Cosmos DB	CDC for Azure SQL MI	CDC for Azure PostgreSQL	CDC for Azure Data Explorer	CDC for SQL Server	CDC for MySQL DB	…
Message & IoT brokers	Azure Service Bus	Solace PubSub+	Google Pub/Sub	MQTT protocol	Azure Event Grid	Azure IoT Hub	…	
Streaming brokers	Apache Kafka	AWS Kinesis Data Streams	AWS MSK	Confluent Cloud	Azure Event Hubs	…		
Events	OneLake events	Azure Events	Fabric events	…				
Feeds	Weather feed	…						

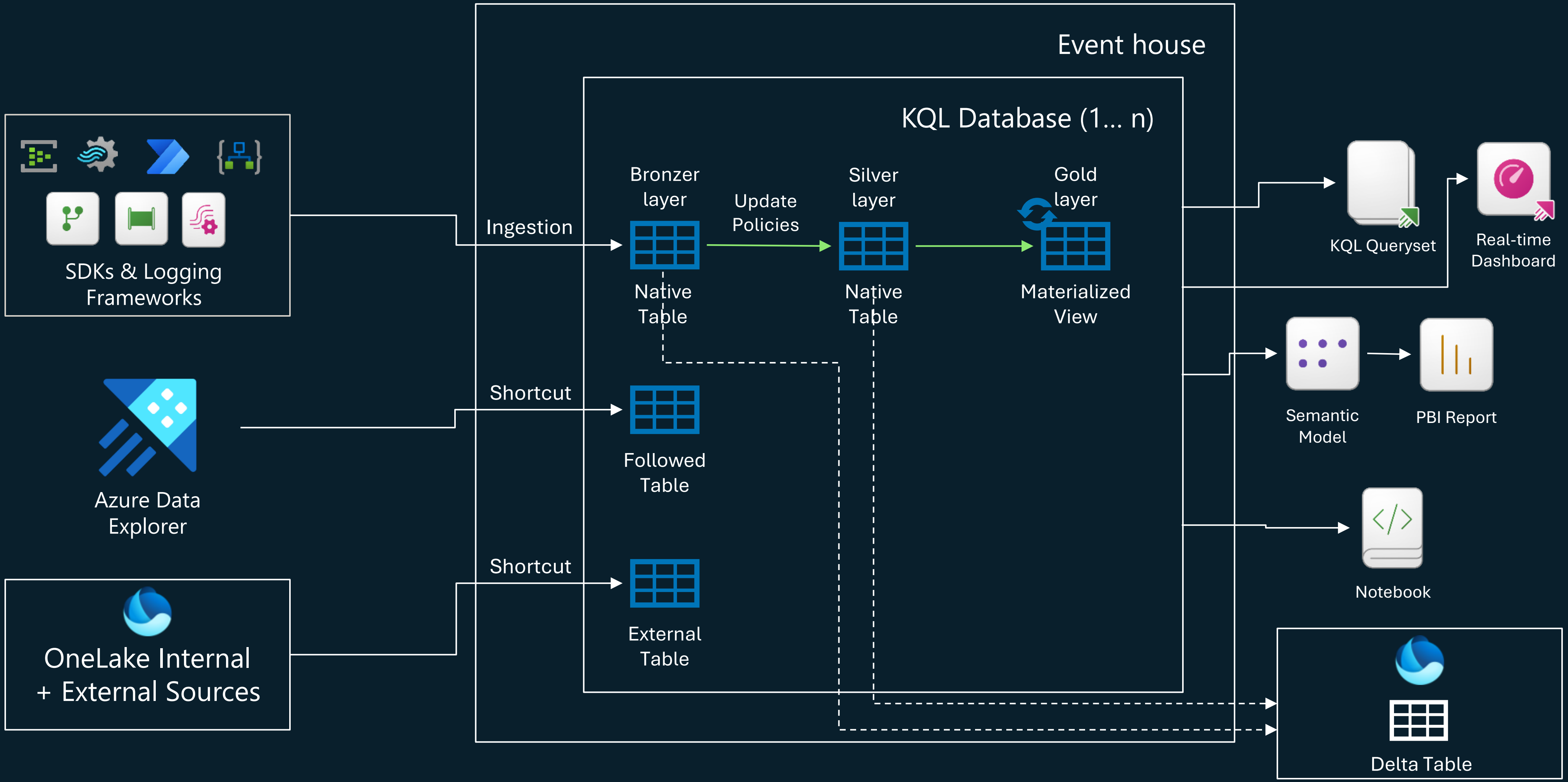
Managed Private Endpoint Enabled

Analyse and transform events using Eventhouse



Workspace Private Link Support,

Event (Real-Time) Medallion Architecture



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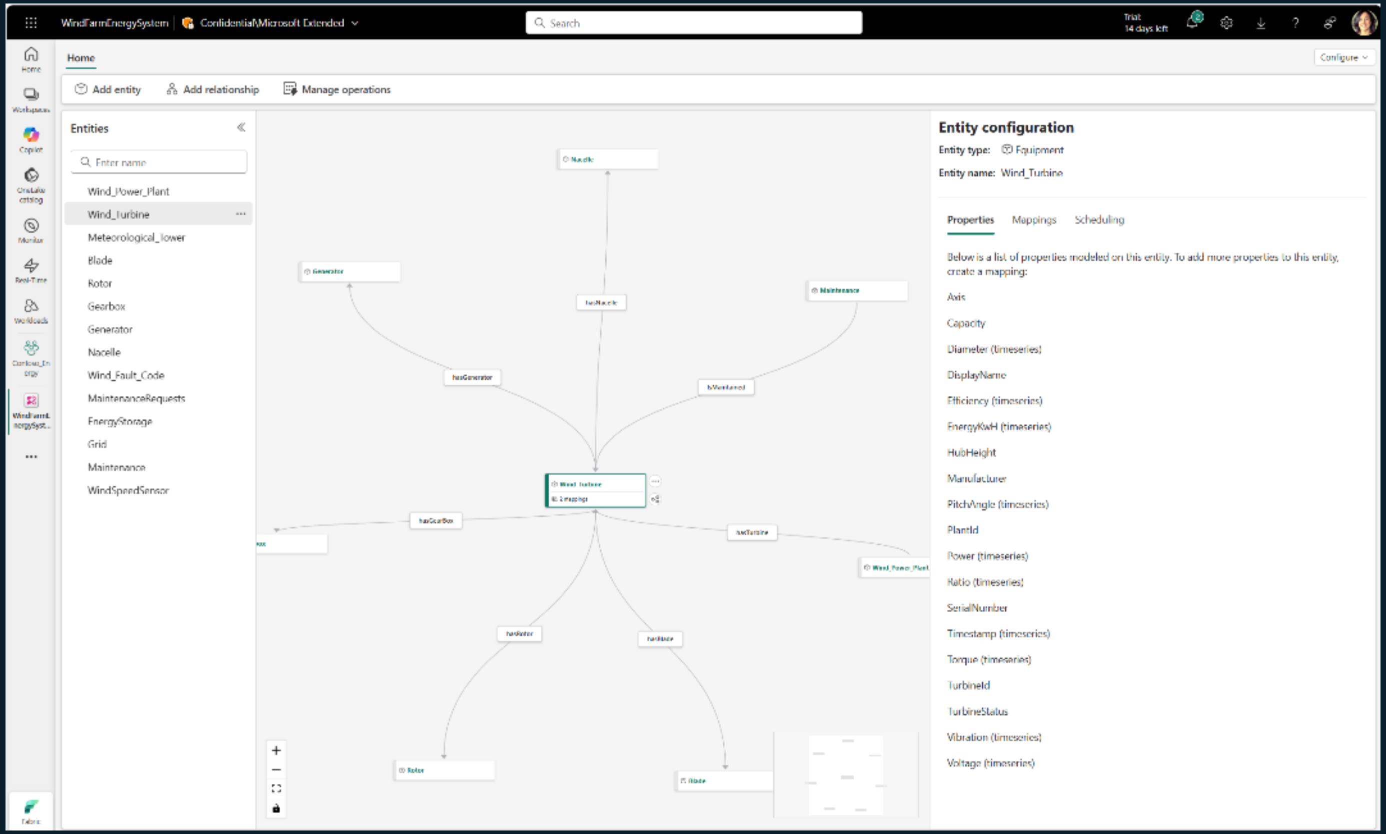
Digital twin builder

Public Preview



Simpler, faster way to build and manage digital twins with a low-code/no-code approach

Easily create and manage ontologies to contextualize data from assets, processes, and systems.

Ontology-powered exploration of complex relationships and uncover contextualized, high-value insights.

Unlock insights with digital twin data in Real-Time Dashboards, Power BI Copilot and Operational Agents

Digital Twin Builder Core Concepts

Namespace

A unique grouping to organize entity types, properties, and relationships.

Example:

Manufacturing Namespace distinguishes elements in a factory domain.

Entity Type

A category defining a concept.

Example:

Centrifugal Pump as a domain-specific entity typecords, rules are evaluated per event

Entity Instance

A specific object of an entity type.

Example:

Pump-001 is an instance of Centrifugal Pump.

Property

A characteristic of an entity type.

Example:

Operating Temperature for a machine

Relationship Type

Defines how entity types are connected.

Example:

hasPart links Pump to Bearing.

Relationship Instance

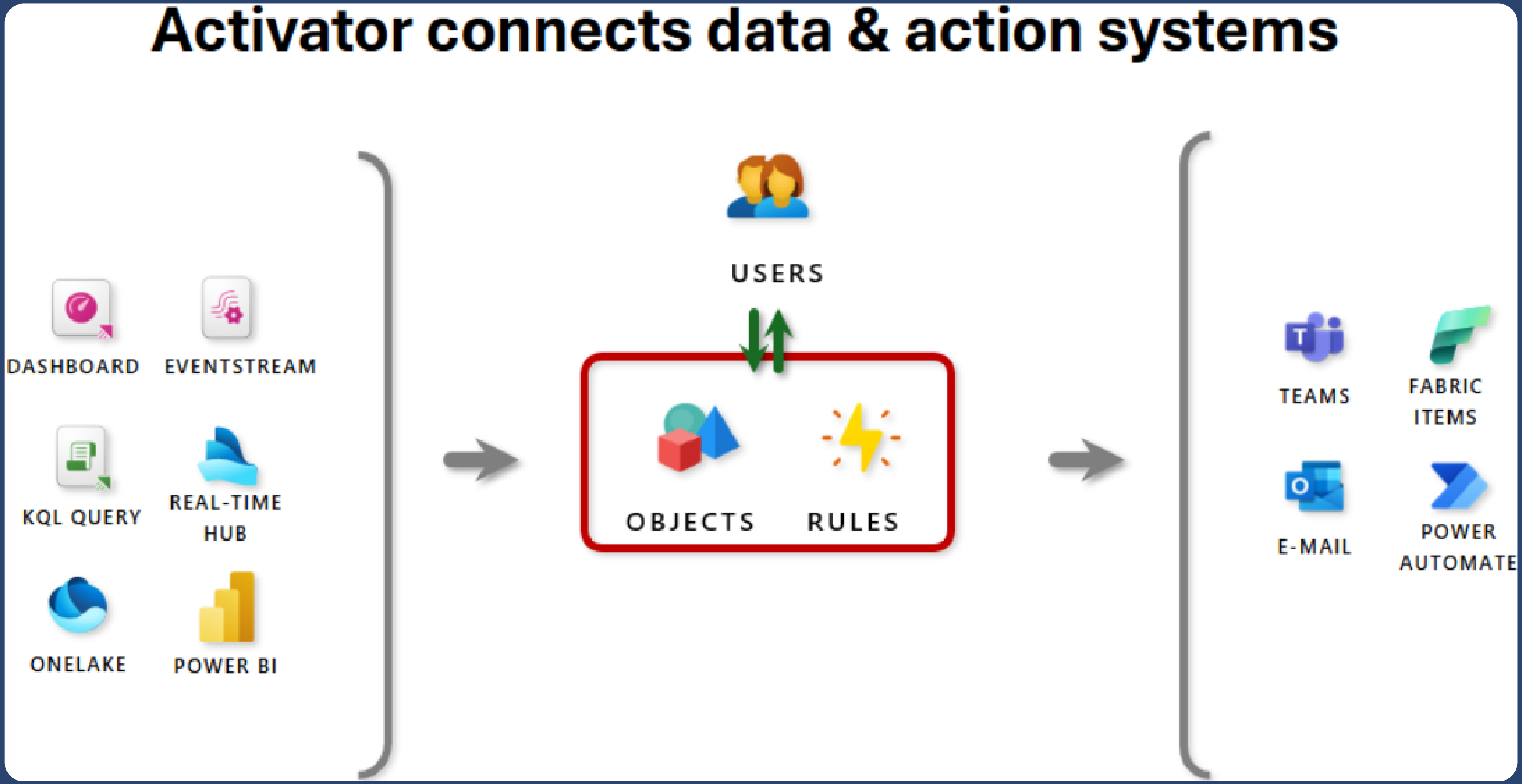
A specific occurrence of a relationship.

Example:

Pump-001 hasPart Bearing-001

https://aka.bi/dtb

Architectural Overview



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Costing

What's behind an Eventstream?

Sources

- Custom Endpoint
- Connected Sources
- Sample Data

Stream

- Under the hood Eventhub

Transformational

(Optional)

- Aggregate
- Expand
- Filter
- Manage Fields
- Union
- Group By
- Join

Under the hood stream analytics jobs

Destinations

- Custom Endpoint
- Derived Stream
- Eventhouse
  - Direct (w/o transformation)
  - Event processing before ingestion
- Lakehouse
- Activator

Eventstream Pricing Model

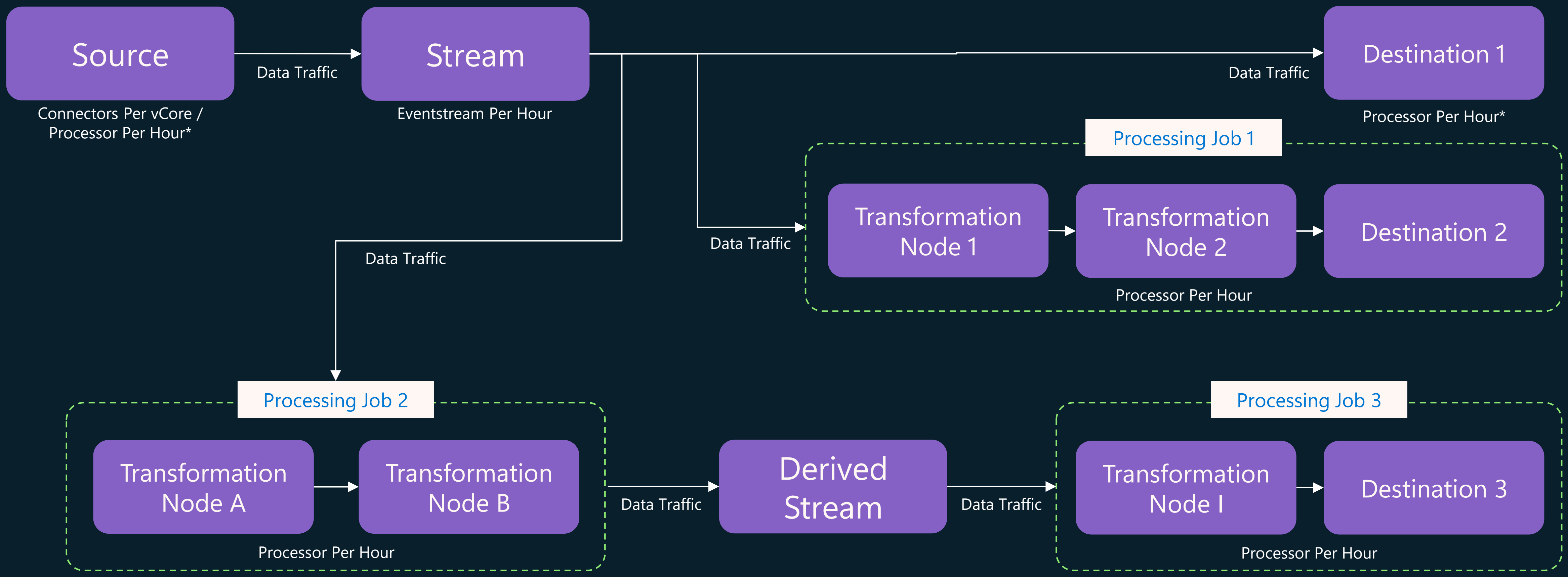
Operation types

Operation	Description
Eventstream Per Hour	A flat charge while the Eventstream is active (event´s been flowing into the stream). If there's no traffic flowing in or out for the past two hours, no charges apply.
Eventstream Data Traffic per GB	Data ingress & egress volume in default and derived streams (Includes 24-hour retention)
Eventstream Processor Per Hour*	Computing resources consumed by the ASA Jobs needed for processing data.
Eventstream Connectors Per vCore Hour**	Computing resources consumed by the connectors.

*CU consumption of the Eventstram Processor Per Hour is designed to correlate with throughput, complexity of processing logic, and partition count of input data. The process autoscale accordingly.
**CU consumption of the Eventstream Connectors Per vCore Hour is designed to correlate with throughput. The process autoscale accordingly.

[Microsoft Fabric event streams capacity consumption - Microsoft Fabric | Microsoft Learn]

How does Eventstream treat CU with multiple routing and transformation paths?



*Custom Endpoints and Eventhouse Direct Ingest don´t contribute to CU consumption

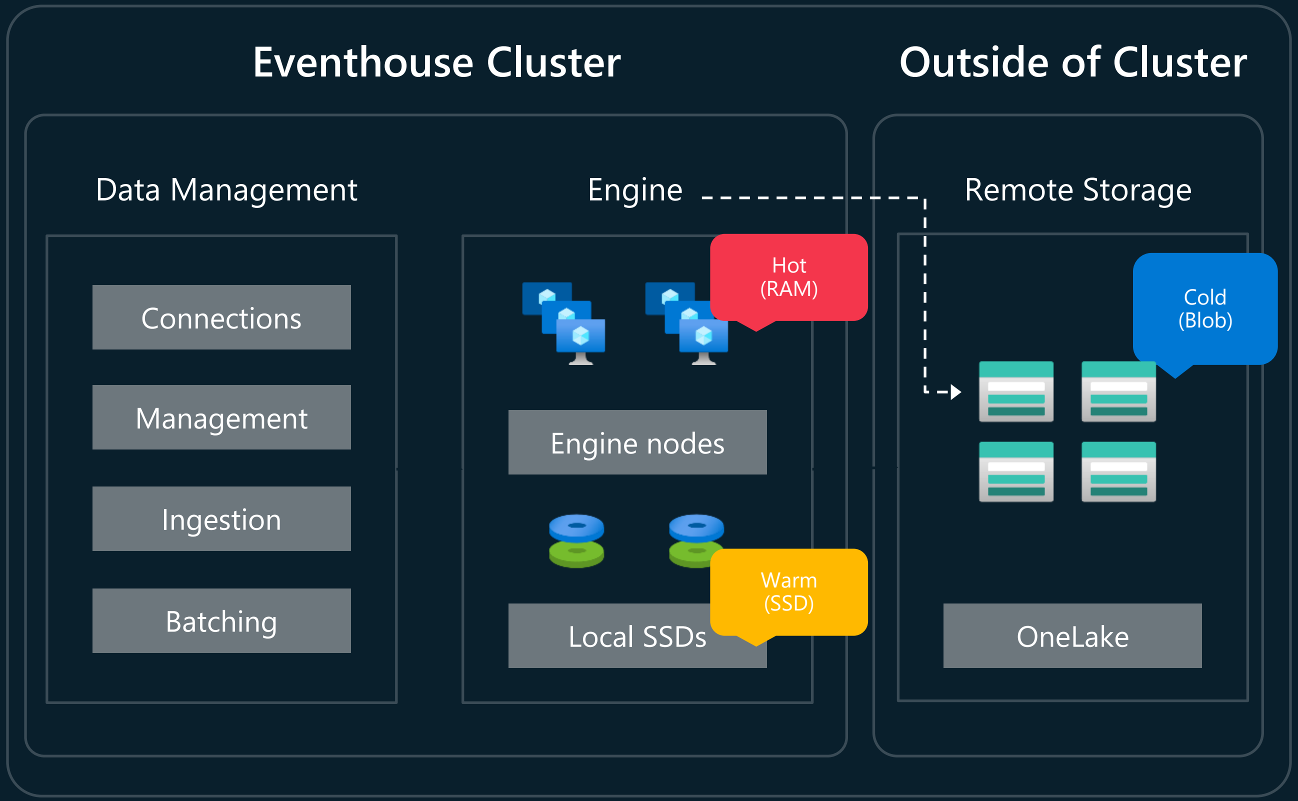
Eventstream Pricing Model

Storage billing

Pricing Principle	Description	
OneLake Standard Storage	Standard storage that's used to persist and store all data. When you set the retention setting for more than 1 day (that is, 24 hours), you're charged as per OneLake Standard storage	

[Microsoft Fabric event streams capacity consumption - Microsoft Fabric | Microsoft Learn]

What´s behind an Eventhouse?



Eventhouse Pricing Model

Eventhouse uses a cluster model for the consumption of CU

Pricing	Description	
Eventhouse UpTime	The number of seconds that your eventhouse is active in relation to the number of virtual cores used by your eventhouse. An autoscale mechanism is used to determine the size of your eventhouse. This mechanism ensures cost and performance optimization based on your usage pattern. An eventhouse with multiple KQL databases attached to it only shows Eventhouse UpTime for the eventhouse item.	
OneLake Cache Storage*	Premium storage that is utilized to provide the fastest query response times. When you set the cache policy, you affect this storage tier.*	
OneLake Standard Storage	Standard storage that is used to persist and store all queryable data. When you set the retention policy, you affect this storage tier.	

*Enabling minimum consumption means that you aren't charged for OneLake Cache Storage. When minimum capacity is set, the eventhouse is always active resulting in 100% Eventhouse UpTime.

Eventhouse and KQL Database consumption - Microsoft Fabric | Microsoft Learn

Share **your thoughts** and help our speakers!

fabfeb.app/feedback